

CS305: Computer Networking

2026 Fall Semester Written Assignment # 1

Due: October 25, 2026, 23:59 (Beijing time)

Please answer questions in English. Using any other language will lead to a zero point.

Q 1. What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?

Q 2. (i) What advantage does a circuit-switched network have over a packet-switched network? What advantages does TDM have over FDM in a circuit-switched network?

(ii) For a P2P file-sharing application, do you agree with the statement, “There is no notion of client and server sides of a communication session”? Why or why not?

Q 3. List the four broad classes of services that a transport protocol can provide. For each of the service classes, indicate if either UDP or TCP (or both) provides such a service.

Q 4. (i) Consider a packet of length L that begins at end system A and travels over three links to a destination end system. These three links are connected by two packet switches. Let d_i , s_i , and R_i denote the length, propagation speed, and the transmission rate of link i , for $i = 1, 2, 3$. The packet switch delays each packet by d_{proc} . Assuming no queuing delays, in terms of d_i , s_i , R_i ($i = 1, 2, 3$), d_{proc} , and L , what is the total end-to-end delay for the packet? Suppose now the packet is 1,500 bytes, the propagation speed on all three links is $2.5 \cdot 10^8$ m/s, the transmission rates of all three links are 2 Mbps, the packet switch processing delay is 3 msec, the length of the first link is 5,000 km, the length of the second link is 4,000 km, and the length of the last link is 1,000 km. For these values, what is the end-to-end delay?

(ii) Equation 1.1¹ gives a formula for the end-to-end delay of sending one packet of length L over N links of transmission rate R . Generalize this formula for sending P such packets back-to-back over the N links.

Q 5.

- (a) Suppose N packets arrive simultaneously to a link at which no packets are currently being transmitted or queued. Each packet is of length L and the link has transmission rate R . What is the average queuing delay for the N packets?
- (b) Now suppose that N such packets arrive to the link every LN/R seconds. What is the average queuing delay of a packet?

¹The referenced equation is $d_{\text{end-end}} = NL/R$.

Q 6. (i) Consider the following string of ASCII characters that were captured by Wireshark when the browser sent an HTTP GET message (i.e., this is the actual content of an HTTP GET message). The characters `<cr><lf>` are carriage return and line-feed characters (that is, the italicized character string `<cr>` in the text below represents the single carriage-return character that was contained at that point in the HTTP header). Answer the following questions, indicating where in the HTTP GET message below you find the answer.

```
GET /cs453/index.html HTTP/1.1<cr><lf>Host: gai
a.cs.umass.edu<cr><lf>User-Agent: Mozilla/5.0 (
Windows;U; Windows NT 5.1; en-US; rv:1.7.2) Gec
ko/20040804 Netscape/7.2 (ax) <cr><lf>Accept:ex
t/xml, application/xml, application/xhtml+xml, text
/html;q=0.9, text/plain;q=0.8, image/png,*/*;q=0.5
<cr><lf>Accept-Language: en-us, en;q=0.5<cr><lf>Accept-
Encoding: zip, deflate<cr><lf>Accept-Charset: ISO
-8859-1, utf-8;q=0.7,*;q=0.7<cr><lf>Keep-Alive: 300<cr>
<lf>Connection:keep-alive<cr><lf><cr><lf>
```

- What is the URL of the document requested by the browser?
- What version of HTTP is the browser running?
- Does the browser request a non-persistent or a persistent connection?
- What is the IP address of the host on which the browser is running?
- What type of browser initiates this message? Why is the browser type needed in an HTTP request message?

(ii) Consider an e-commerce site that wants to keep a purchase record for each of its customers. Describe how this can be done with cookies.

Q 7. Consider Figure 2.12, for which there is an institutional network connected to the Internet. Suppose that the average object size is 850,000 bits and that the average request rate from the institution's browsers to the origin servers is 16 requests per second. Also suppose that the amount of time it takes from when the router on the Internet side of the access link forwards an HTTP request until it receives the response is three seconds on average (see Section 2.2.5). Model the total average response time as the sum of the average access delay (that is, the delay from Internet router to institution router) and the average Internet delay. For the average access delay, use $\Delta/(1 - \Delta\beta)$, where Δ is the average time required to send an object over the access link and β is the arrival rate of objects to the access link.

- Find the total average response time.
- Now suppose a cache is installed in the institutional LAN. Suppose the miss rate is 0.4. Find the total response time.

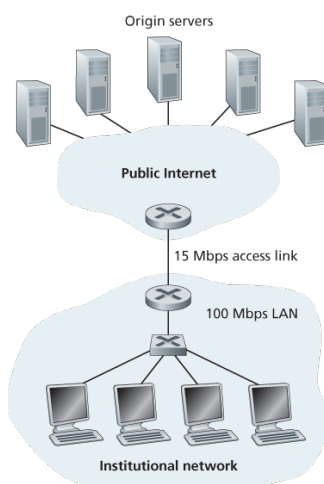


Figure 2.12.

Q 8. (i) Suppose within your Web browser you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your local host, so a DNS lookup is necessary to obtain the IP address. Suppose that n DNS servers are visited before your host receives the IP address from DNS; the successive visits incur an RTT of $RTT_1, \dots, RTT_n$. Further suppose that the Web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Let RTT_0 denote the RTT between the local host and the server containing the object. Assuming zero transmission time of the object, how much time elapses from when the client clicks on the link until the client receives the object?

(ii) Referring to Problem P7², suppose the HTML file references eight very small objects on the same server. Neglecting transmission times, how much time elapses with

- (a) Non-persistent HTTP with no parallel TCP connections?
- (b) Non-persistent HTTP with the browser configured for 5 parallel connections?
- (c) Persistent HTTP?

Q 9. Consider distributing a file of F bits to N peers using a client-server architecture. Assume a fluid model where the server can simultaneously transmit to multiple peers, transmitting to each peer at different rates, as long as the combined rate does not exceed u_s .

- (a) Suppose that $u_s/N \leq d_{\min}$. Specify a distribution scheme that has a distribution time of NF/u_s .
- (b) Suppose that $u_s/N \geq d_{\min}$. Specify a distribution scheme that has a distribution time of $F/d_{\min}$.
- (c) Conclude that the minimum distribution time is in general given by $\max\{NF/u_s, F/d_{\min}\}$.

²Problem P7 is reproduced in Q8(i).